

## HARDWARE REVIEW

### Spectravideo

### SV-318

# One nifty feature after another

By Martin Downey

When I went to collect the Spectravideo computer for review I had preconceptions. There are so many home computers around already that my first impression when I saw the glossy cardboard box was, "Not another one!" But once out of the box the computer started to take on a new appearance as I discovered one nifty feature after another.

The computer case is white plastic with keys of grey with white markings. Although the keys are rubber they are in the full size QWERTY layout. There is a long space-bar, two shift keys and a caps-lock key with LED. There are two graphic symbols above most keys and these are accessed using the LEFT-GRAPH and RIGHT-GRAPH keys.

There are 10 programmable functions keys (five physical keys used with SHIFT). On the right of the keyboard is the first "nifty-feature", an eight-way cursor pad which doubles as a joystick by inserting a red knob (a spare knob is also supplied). This cursor pad is used in association with these other keys (called "Wordprocessing keys") to give very powerful, easy-to-use full-screen editing of BASIC programs. This is a considerable improvement on the standard Microsoft EDIT command. The three editing keys are CLS/HOME (home cursor and/or clear screen), INS/PASTE (insert mode on/off) and DEL/CUT (delete character). A fourth key SELECT is not mentioned in the documentation, but its close proximity to the cursor pad suggests a "mouse-like" use given appropriate software (shades of the Apple LISA).

Both the power supply and VHF modulator are external to the computer and there is also a



*The Spectravideo SV-318*

TV/Computer switch box. All this makes for quite a jumble and makes the system much less portable than the glossies imply.

The back of the computer reveals an expansion socket and the cassette socket. The Spectravideo cannot use a standard recorder so this means an additional cost of \$129. However, the special recorder will load and save programs at 1800 baud, which is quite a bit faster than can be achieved with the average home cassette recorder. I did not get a chance to review the recorder, so I cannot comment on reliability.

On the right of the computer is the power socket and on/off switch as well as two joystick ports. These are standard sockets so most brands of joysticks should be compatible. The sockets are also used for an optional graphics tablet, which at \$199 looks worth-while. However, details of the tablet were scarce and its usefulness would depend on appropriate software being available.

At the top right of the computer is

the games cartridge socket, which is nicely concealed under a hinged door. Prices for cartridges range from \$18 to \$40, which is quite good value when you consider that cassette-based software from some other computers costs just as much. The advantages of ROM cartridges cannot be overstated. Instant loading, no degradation (unless you violently misuse them), and virtually no home pirating. I had the opportunity to try out two such game cartridges: Frantic Freddy and Super Cross-Force. Frantic Freddy is a brilliantly addictive game with great graphics. The scenario is an old-apartment block (complete with graffiti!) with you as the fireman trying to extinguish the flames and rescue cats. Great fun and ideal for "kids" of all ages. Super Cross-Force is less spectacular, but is still a challenging Invaders-style game for those not spoilt on more elaborate computer games.

If you have a cartridge in place when the computer is switched on

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then that program will automatically be run otherwise the computer will come up in Microsoft BASIC. It is this BASIC that really sets the Spectravideo apart from its competitors.

The first noticeable feature of the BASIC Operating System is the automatic self-test at power-up. This takes just a fraction of a second and then the company logo appears. After this you are into extended Microsoft BASIC. This is the version of BASIC found on most CP/M and MP/M machines with additional commands for sound and graphics.

### POWERFUL FUNCTION-KEY SYSTEM

The computer powers up in screen mode 0 which is 40 x 23 text. The default colours are white-on-blue, which gives a surprisingly readable display even on a poorly adjusted TV. I cannot understand why some other home computers still make do with just 21 or 32 column text. The colour scheme can be changed to any combination of the 16 available colours.

There is also a twenty-fourth line in this screen mode which shows the function-key definitions.

This line is automatically changed to show the shifted definitions every time SHIFT is pressed. The keys are loaded with default definitions such as RUN and LIST when the computer is turned on, but are easily changed to any 15-character ASCII string using the KEY commands.

Altogether this is a very powerful system usually found only in up-market business machines and it certainly outshines the use of function keys on other home computers. Another BASIC command, ON KEY GOSUB, which works under interrupt, adds still more to the computer's power.

Other commands of note are:

- ON STRIG GOSUB, which branches whenever the joystick fire button or space-bar is pressed.
- ON SPRITE GOSUB, which branches whenever two sprites (defined pictures) collide.
- PLAY, which plays a tune using a Music Macro Language giving notes A-G sharps, flats, change octaves, tempo, volume...
- DRAW, which draws pictures using a Graphic Macro Language with commands for up, down, left, right, angle, diagonal, colour, scaling...
- SOUND, which controls the four

## Microcomputer summary

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name:</b>               | Spectravideo SV-318                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Manufacturer:</b>       | Spectravideo, U.S.A. (Hardware from Hong Kong, Software from Tokyo division of Microsoft).                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Microprocessor:</b>     | Z80A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Clock Speed:</b>        | 3.6 MHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>RAM:</b>                | 16K video + 16K user built in. Expandable to 144K externally (bank switched)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>ROM:</b>                | 32K Microsoft BASIC built in. Expandable to 96K externally.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Input-Output:</b>       | Expansion slot. Cassette port (non-standard). 2 x joystick ports (also used by Graphics Tablet). Game cartridge socket.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Keyboard:</b>           | 54-key standard QWERTY, 10 programmable (5 physical keys), 2 graphic function, 4 editing, 8 way joystick/cursor-keypad, STOP key. Rubberised keys.                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Display:</b>            | Composite video output through VHF modulator to standard TV (Channel 2). Includes sound.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Languages:</b>          | Microsoft Extended BASIC built in. Disk BASIC and CP/M options available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Graphics:</b>           | SCREEN 0 — 40 x 23 (plus function-key line) text & graphic characters only, 2 colour.<br>SCREEN 1 — 256 x 192 graphics, 16 colours, 40 x 24 text.<br>SCREEN 2 — 64 x 48 graphics, 16 colours, 11 x 6 text.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Sound:</b>              | 3 channels of music (7 octaves) plus 1 channel noise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Price:</b>              | \$899.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Options:</b>            | SV-328 Business version (\$1475 total).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Peripherals:</b>        | Super expander (\$359), mini expander (\$41)<br>Games adapter — runs COLECO games (\$216)<br>Dot-matrix printer with interface card (\$1245)<br>Floppy-disk drive — 173K (\$995)<br>Cassette drive (\$129), graphics tablet (\$199)<br>External joysticks (\$27.50 each)<br>Interface cartridge for expander units:<br>Disk drive cartridge, CP/M + Disk BASIC (\$331)<br>Centronics interface (\$193)<br>RS232 interface (\$193)<br>80 column card (\$345), modem card (\$322)<br>16K RAM card (\$106), 64K RAM card (\$350)<br>Games cartridges \$18-\$40 each. |
| <b>Other features:</b>     | Out of 5. Documentation 3, ease of use 5, language 5, expansion 4, value for money 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Reviewer's ratings:</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

*Review Unit From: Computer South, 78 Oxford Terrace, Christchurch*

sound channels including envelope information.

- CIRCLE, LINE, PAINT, POINT, GET, PSET, PRESET.

The list goes on, but one command I must give special mention to, because I have not seen it in any other version of BASIC, is ON INTERVAL =n GOSUB. This command lets you specify a routine that is executed every n/60 seconds. Such programmable interrupts are usually available only through machine code. By making them available from BASIC the door is opened to all sorts of industrial and home-control applications. Brilliant!

Another command of awesome power for a machine its price is SWITCH. This lets you switch

between banks of memory if you have the optional extra RAM installed. Used in conjunction with the interrupts "multi-tasking" is possible (running two or more programs concurrently). "Are you sure this is just a home computer?"

I have already mentioned screen mode 0 which is just text and graphics characters (similar to PET graphics). SCREEN 1 is true high-resolution graphics with 16 colours. SCREEN 2 is low-resolution graphics with 16 colours. The low-resolution mode of 64 x 48 has very limited applications, but would probably be useful for those just learning about computer graphics. A medium resolution of 128 x 96 would have been more useful.



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Screen mode 2 with 256 x 192 pixels is the optimum resolution from a programmer's point of view. Higher resolutions are nice to have, but seldom get used because of the extra programming involved.

On the surface the documentation appears very good. It comes in a glossy ring-binder and includes lots of pictures. The sections on setting up the computer and the introduction to BASIC are very good, but there is not enough explanation of the more advanced commands. I had to find out about a number of the commands through trial and error. A removable booklet in the back of the ring-binder acts as a quick-reference guide to BASIC, but some of the commands are never explained in detail at all. There is certainly room for follow-up material. The English used is good (certainly not Asian-English), but there are too many frustrating typographical errors.

The SV-318 can certainly be expanded well beyond most other home computers and since all additions are just "plug in" they are ideal for the novice. However, such expansion is generally more expensive than buying all-in-one so if you want many of the options, one

of the higher-priced computers may be cheaper in the long run.

A Business version (SV-328) is also available although one was not yet available here for review. This has a proper 87-key typewriter keyboard including cursor keys and a numeric keypad. It also has 48K ROM and 80K RAM built in and at \$1475 with the option of running CP/M it is a very competitive business machine (plus it will run the SV-318 games!).

### Conclusions

- Good colour and sound.
- An excellent version of BASIC.
- Ample expansion options.
- Lots of nice little features that make the computer fun to use.

### Sample listings of Spectravideo BASIC.

#### LISTING 1: Use of interrupts

```
10 INTERVAL ON
20 ON INTERVAL=10 GOSUB 100
30 FOR I=1 TO 3000
40 PRINT I;
50 NEXT I
60 END
100 PRINT "****";
110 RETURN
```

#### LISTING 2: Use of sprites and sprite collisions

```
110 SCREEN 1: 'High resolution
120 ON SPRITE GOSUB 240: 'For collision
130 SPRITE ON
132 '
135 'Read in picture of ball for sprite
140 FOR T=1 TO 8
150 READ A$
160 S$=S$+CHR$(VAL("&B"+A$))
170 NEXT T
180 SPRITE $(1)=S$
182 '
185 'Move 2 sprites till they collide
190 FOR I=1 TO 256
200 PUT SPRITE 0, (256-I, 96), 8, 1
210 PUT SPRITE 1, (I, 96), 7, 1
220 NEXT I
230 PRINT "MISS": GOTO 230
235 '
240 PRINT "HIT": GOTO 240
245 '
250 DATA 00011000
260 DATA 00111100
270 DATA 01111110
280 DATA 01111110
290 DATA 01111110
300 DATA 01111110
310 DATA 00111100
320 DATA 00011000
```

#### LISTING 3: Use of DRAW and PLAY

```
10 SCREEN 1
20 BOX$="C8U50R50C7D50L50"
30 DRAW BOX$
40 '
50 HAPPY$="S5LBC64R64C64DCEF"
60 PLAY HAPPY$
```

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